

## Lecture 3 | Lecturer : Li Yu | Wednesday , 16 September 2026

### Section1 : Sequence data

Reason for sequence data :

Central dogma -> **genetic information** is hidden in DNA sequence

Phenotype = genotype + Environment

Example of Phenotype : height , weight , HbA1c, eye color etc.

DNA sequence :

category : A,C,T,G -> Base

structure : Double stranded

Table for DNA/RNA/Protein :

|           | DNA             | RNA              | Protein              |
|-----------|-----------------|------------------|----------------------|
| category  | A,C,T,G         | A,C, <b>U</b> ,G | 20 amino acid        |
| structure | double stranded | single stranded  | 3D folding structure |
| method    | sequencing      | sequencing       | mass spectrometry    |

How DNA/RNA sequencing evolve ?

- short read -> long reads
- lower error rate

The latest sequencing technology : nanopore sequencing

method :

1. DNA enter a chemical pore
2. different base( A,C,T,G ) will have electrical current change
3. detecting the change -> obtain the sequence

pros: longer read

cons : error rate higher

mass spectrometry :

step 1 : break long sequence to short pieces

step 2 : each piece determined by **Mass Spectrometry**

step 3 : get the **weight** of each piece

step 4 : **assembly** the piece into sequence

What will do to the DNA sequencing raw data ?

Step 1 : sequencing ( WGS, WES ) -> obtain FASTQ data

Step 2: quality control -> mapping

step 3 : **variant** calling and analysis

Sequence-to-Structure-to-Function Paradigm :

1. similar **sequence** -> similar **structure** -> similar **function**

2. similar sequence -> common ancestor  
 -> compare the sequence = determine the **evolution** and **function** from the sequence

## Section 2 : Sequence comparison and alignment score

Pairwise sequence alignment : arrange 2 sequence to maximize similarity

method to define sequence similarity :

A <-> A = MATCH , score = 2

G <-> T = Mismatch score = -7

A <-> G , score = -5

G <-> \_ = Gap , score = -10

\*\*\* The score matrix is just a **reference**, it can **change** and the **outcome will be affected**

method to find the best pairwise alignment :

enumeration :

CALCULATE ALL SCORES

select the **highest**

Limit : even n = 300, possible alignment =  $7 * 1088$  which is larger than no. of particles in visible universe

So we need **dynamic programming**

## Section 3 : Dynamic programming

step1 : break problem -> sub-problem

step 2 : solve sub-problems **optimally** and **recursively**

optimally : choose the best score

recursively : solve the **subproblem repeatedly** using the same logic/method

step3 : use those **optimal** solutions -> optimal solutions for the original problem

Real Life Example : shortest path from CUHK MTR station to Science Center

(station , YIA ) = 3 min , (station, Ponn) = 5 min , (YIA, HSB ) = 8 min , (HSB, SC ) = 7 min

(Ponn , Shaw SCI ) = 12 min , (Shaw SCI , SC ) = 5 min

Path 1                      Shortest                      +                      Shortest                      +                      Shortest  
                                          (station , YIA )                      (YIA , HSB )                      (HSB, SC )

Shortest:  
 (station , SC )

Path 2                      Shortest                      +                      Shortest                      +                      Shortest  
                                          (station , Ponn )                      (Ponn, Shaw SCI )                      (Shaw SCI, SC )

path 1 =  $3 + 8 + 7 = 18$  min

path 2 =  $5 + 12 + 5 = 22$  min

Hence, path 1 is optimal path for walking from MTR station to Science Center

More example about using ACTG base of DNA

## Example 1 — TGA vs AGCA

**Original scoring matrix** (match = +2, A ↔ G / C ↔ T = -5, other = -7, gap = -10)

|   | A  | C  | G  | T  |
|---|----|----|----|----|
| A | 2  | -7 | -5 | -7 |
| C | -7 | 2  | -7 | -5 |
| G | -5 | -7 | 2  | -7 |
| T | -7 | -5 | -7 | 2  |

|   |     | A   | G   | C   | A   |
|---|-----|-----|-----|-----|-----|
|   | 0   | -10 | -20 | -30 | -40 |
| T | -10 | -7  | -17 | -25 | -35 |
| G | -20 | -15 | -5  | -15 | -25 |
| A | -30 | -18 | -15 | -12 | -13 |

Best alignment:  $\begin{matrix} \text{T} & \text{G} & - & \text{A} \\ \text{A} & \text{G} & \text{C} & \text{A} \end{matrix}$  **score = -13**

Comparison: the best alignment is TG-A / AGCA under both matrices — only the score changes (-13 → -3).

**New scoring matrix** (match = +1, transition = -1, transversion = -3, gap = -2)

|   | A  | C  | G  | T  |
|---|----|----|----|----|
| A | 1  | -3 | -1 | -3 |
| C | -3 | 1  | -3 | -1 |
| G | -1 | -3 | 1  | -3 |
| T | -3 | -1 | -3 | 1  |

|   |    | A  | G  | C  | A  |
|---|----|----|----|----|----|
|   | 0  | -2 | -4 | -6 | -8 |
| T | -2 | -3 | -5 | -5 | -7 |
| G | -4 | -3 | -2 | -4 | -6 |
| A | -6 | -3 | -4 | -5 | -3 |

Best alignment:  $\begin{matrix} \text{T} & \text{G} & - & \text{A} \\ \text{A} & \text{G} & \text{C} & \text{A} \end{matrix}$  **score = -3**

## Example 2 — ACGGTCGATC vs CCTGTCCATG

**New scoring matrix** (match = +1, everything else = 0, gap = 0)

|   | A | C | G | T |
|---|---|---|---|---|
| A | 1 | 0 | 0 | 0 |
| C | 0 | 1 | 0 | 0 |
| G | 0 | 0 | 1 | 0 |
| T | 0 | 0 | 0 | 1 |

|   |   | C | C | T | G | T | C | C | A | T | G |
|---|---|---|---|---|---|---|---|---|---|---|---|
|   | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| A | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| C | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| G | 0 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| G | 0 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 3 |
| T | 0 | 1 | 1 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 |
| C | 0 | 1 | 2 | 2 | 2 | 3 | 4 | 4 | 4 | 4 | 4 |
| G | 0 | 1 | 2 | 2 | 3 | 3 | 4 | 4 | 4 | 4 | 5 |
| A | 0 | 1 | 2 | 2 | 3 | 3 | 4 | 4 | 5 | 5 | 5 |
| T | 0 | 1 | 2 | 3 | 3 | 4 | 4 | 4 | 5 | 6 | 6 |
| C | 0 | 1 | 2 | 3 | 3 | 4 | 5 | 5 | 5 | 6 | 6 |

Best alignment:  $\begin{matrix} \text{A} & \text{C} & \text{G} & \text{G} & \text{T} & \text{C} & \text{G} & \text{A} & \text{T} & \text{C} \\ \text{C} & \text{C} & \text{T} & \text{G} & \text{T} & \text{C} & \text{C} & \text{A} & \text{T} & \text{G} \end{matrix}$  **score = 6**

RNA also use dynamic programming ( RNA uses A, C, **U**, G, U replaces T )

**Example 1: AUGCA vs AGCA**

|   | A  | U  | C  | G  |
|---|----|----|----|----|
| A | 2  | -7 | -7 | -5 |
| U | -7 | 2  | -5 | -7 |
| C | -7 | -5 | 2  | -7 |
| G | -5 | -7 | -7 | 2  |

Best alignment:

|   |   |   |   |   |
|---|---|---|---|---|
| A | U | G | C | A |
| A | - | G | C | A |

score = -2

Gap penalty = -10

|   |     | A   | G   | C   | A   |
|---|-----|-----|-----|-----|-----|
|   | 0   | -10 | -20 | -30 | -40 |
| A | -10 | 2   | -8  | -18 | -28 |
| U | -20 | -8  | -5  | -13 | -23 |
| G | -30 | -18 | -6  | -12 | -18 |
| C | -40 | -28 | -16 | -4  | -14 |
| A | -50 | -38 | -26 | -14 | -2  |

**Example 2: ACGU vs ACG**

|   | A  | U  | C  | G  |
|---|----|----|----|----|
| A | 2  | -7 | -7 | -5 |
| U | -7 | 2  | -5 | -7 |
| C | -7 | -5 | 2  | -7 |
| G | -5 | -7 | -7 | 2  |

Best alignment:

|   |   |   |   |
|---|---|---|---|
| A | C | G | U |
| A | C | G | - |

score = -4

Gap penalty = -10

|   |     | A   | C   | G   |
|---|-----|-----|-----|-----|
|   | 0   | -10 | -20 | -30 |
| A | -10 | 2   | -8  | -18 |
| C | -20 | -8  | 4   | -6  |
| G | -30 | -18 | -6  | 6   |
| U | -40 | -28 | -16 | -4  |